



Caringbah High School

2019

Trial HSC Examination

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen only
(Black pen is preferred)
- NESA approved calculators
may be used
- A reference sheet is provided
- In Questions 11–16, show
relevant mathematical
reasoning and/or calculations

Total marks – 100

Section I 10 marks

Pages 2 – 5

- Attempt Questions 1–10
- Allow about 15 minutes for this
section

Section II 90 marks

Pages 6 – 14

- Attempt Questions 11–16
- Allow about 2 hour and 45
minutes for this section

Question 1 - 10 (1 mark each) Answer on page provided.

1 Let $z = 1 + 2i$ and $\omega = 3i - 4$. What is the value of $z\bar{\omega}$?

A) $4 + i$

B) $2 - 3i$

C) $2 - 11i$

D) $-2 + 11i$

2 Let A, B and C be three consecutive terms of an arithmetic sequence.

Which of the following is a simplification of $\frac{\sin(A + C)}{\sin B}$?

A) $2 \cos B$

B) $\sin 2B$

C) $\cot B$

D) 2

3 The eccentricity of the ellipse $\frac{x^2}{100} + \frac{y^2}{36} = 1$ is $e = \frac{4}{5}$.

Which of the following is the distance between the two foci?

A) 8

B) 16

C) 20

D) 25

4 The polynomial equation $x^3 - 3x^2 + 2 = 0$ has roots α, β and γ .

What is the value of $\alpha^3 + \beta^3 + \gamma^3$?

A) 9

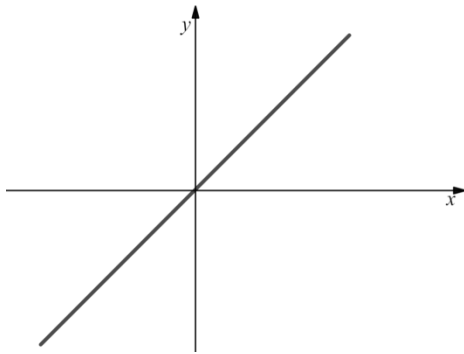
B) 13

C) 21

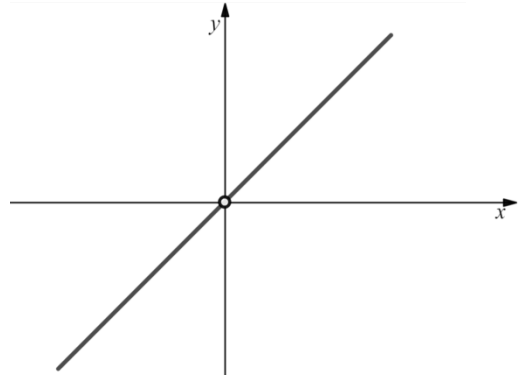
D) 25

- 5 The equation $\frac{x}{y} + \frac{y}{x} = 2$ defines y implicitly as a function of x .
Which of the following graphs best represents this implicit function?

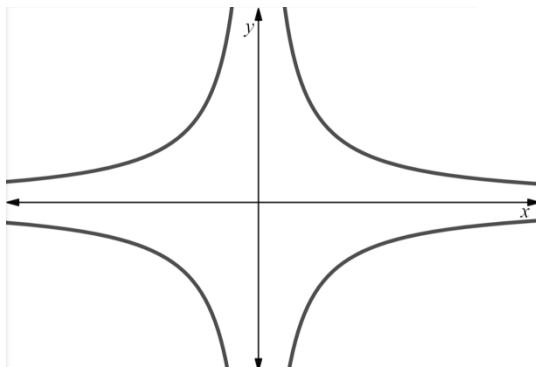
A)



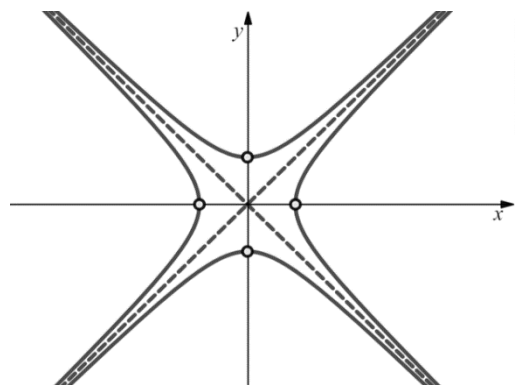
B)



C)



D)



- 6 If $x = 1 - \cos \theta$ and $y = \theta - \sin \theta$, which is the result for $\frac{dy}{dx}$?

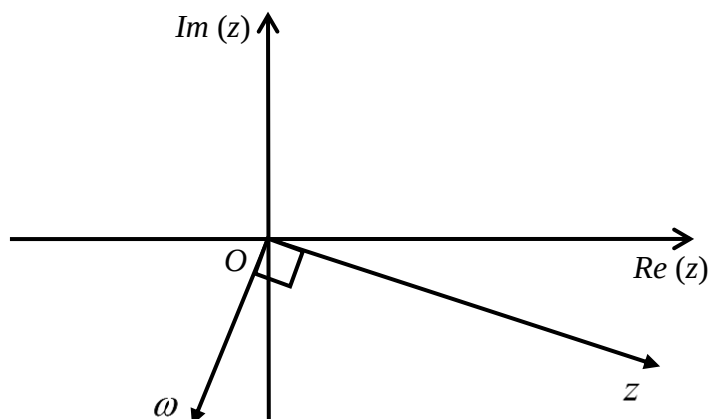
A) $\frac{dy}{dx} = -\cot \frac{\theta}{2}$

B) $\frac{dy}{dx} = -\tan \frac{\theta}{2}$

C) $\frac{dy}{dx} = \cot \frac{\theta}{2}$

D) $\frac{dy}{dx} = \tan \frac{\theta}{2}$

- 7 Consider the following diagram, drawn to scale, showing the complex numbers z and ω .



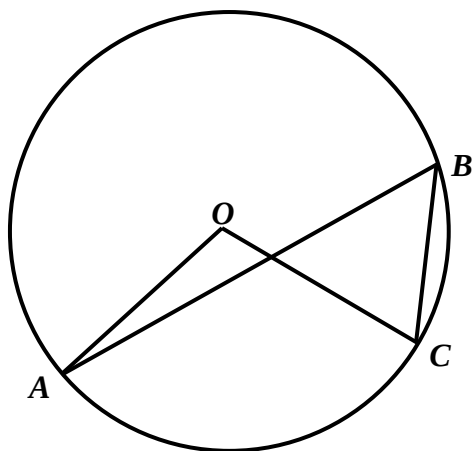
Which of statements below is false?

- A) $|z + \omega| = |z - \omega|$ B) $\operatorname{Re}\left(\frac{\omega}{z}\right) = 0$
- C) $z^2 = k\omega^2$, where k is real D) $-\pi < \arg(z - \omega) < 0$
- 8 Which of the following is the domain of the function $y = \cos^{-1}\left(\frac{x-a}{b}\right)$, $b > 0$?

- A) $a - b \leq x \leq a + b$ B) $a - 1 \leq x \leq a + 1$
- C) $a \leq x \leq a + b\pi$ D) $-1 \leq x \leq 1$

9 In the diagram below, O is the centre of the circle.

It is given that $\angle OAB = 20^\circ$ and $\angle OCB = 52^\circ$.



What is the size of $\angle ABC$?

- A) 49°
- B) 32°
- C) 56°
- D) 64°

10 A particle is moving in a straight line with velocity at any particular time given by $v = \sin^{-1} x$.

Which of the following represents the acceleration of the particle?

- A) $\frac{\sin^{-1} x}{\sqrt{1-x^2}}$

B) $\frac{-\sin^{-1} x}{\sqrt{1-x^2}}$

C) $\frac{1}{\sqrt{1-x^2}}$

D) $\frac{-\cos x}{\sin^2 x}$

END OF MULTIPLE CHOICE QUESTIONS

Section II (90 marks)**Attempt all questions 11–16****Allow about 2 hours and 45 minutes for this section**

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Start a NEW booklet.**Marks**(a) Let $z = \sqrt{3} + i$.(i) Express z in modulus – argument form.

2

(ii) Hence find z^4 in $x + iy$ form.

2

(b) A point $P(z)$ moving in the complex plane has its locus in terms of z defined by $|z - 1| = |z + 2 - 3i|$.

Find the cartesian equation of the locus.

2

(c) Find $\int \sin \theta \cos^5 \theta \, d\theta$.

2

(d) (i) Find the values of A , B and C so that

2

$$\frac{5}{(x+1)(x^2+4)} \equiv \frac{A}{x+1} + \frac{Bx+C}{x^2+4}.$$

(ii) Hence find $\int \frac{5}{(x+1)(x^2+4)} \, dx$.

3

(e) Consider the polynomial $P(x) = x^3 + 3x^2 - 24x + k$.If $P(x) = 0$ has a zero of multiplicity 2, find all possible values of k .

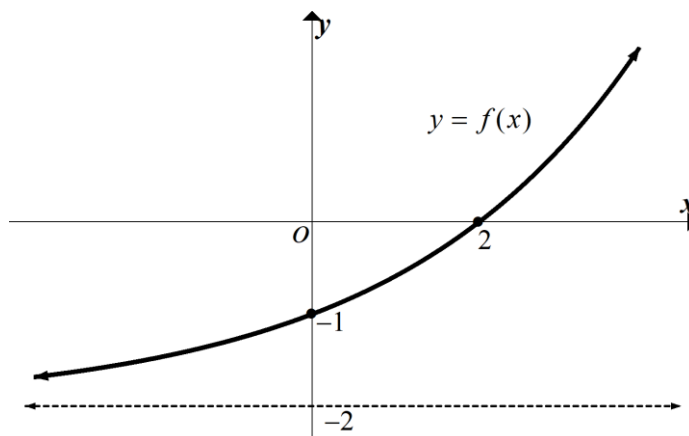
2

Question 12 (15 marks) Start a NEW booklet.**Marks**

(a) Find $\int \frac{x^2}{x+1} dx$.

2

(b) The diagram below shows the graph of $y = f(x)$.



On separate sketches, showing any important features, neatly draw the graphs of

(i) $y = f(-x)$

1

(ii) $y = \sqrt{f(x)}$

1

(iii) $y = [f(x)]^2$

2

(iv) $y = \frac{1}{f(x)}$

2

(v) $y = \log_e [f(x)]$

2

(vi) $y = 1 + \frac{d}{dx} [f(x)]$

2

Question 12 continues on page 8

Question 12 (continued)

Marks

- (c) On the Argand diagram shade the region defined by

3

$$|z - i| \leq 2 \text{ and } 0 \leq \arg(z - 1) \leq \frac{3\pi}{4}.$$

End of Question 12

Question 13 (15 marks) Start a NEW booklet.**Marks**(a) Consider the hyperbola $x^2 - 9y^2 = 1$. Find:

- | | | |
|-------|----------------------------------|----------|
| (i) | the eccentricity. | 1 |
| (ii) | the coordinates of the foci. | 1 |
| (iii) | the equation of the directrices. | 1 |

(b) Consider the ellipse $E: \frac{x^2}{9} + \frac{y^2}{4} = 1$.

Normals to E at the points $P(3\cos\alpha, 2\sin\alpha)$ and $Q(3\cos\beta, 2\sin\beta)$ are at right angles to each other.

- | | | |
|------|--|----------|
| (i) | Show that the gradient of the normal at P is $\frac{3\sin\alpha}{2\cos\alpha}$. | 2 |
| (ii) | Hence or otherwise, find the value of $\cot\alpha \cot\beta$. | 2 |

c) If α, β and γ are the roots of the equation $x^3 + 6x + 1 = 0$, find in simplest form the polynomial equation with roots $\alpha\beta, \beta\gamma$ and $\alpha\gamma$.

3

d) Consider the function $f(x) = x - \ln(1 + x^2)$.

- | | | |
|------|--|----------|
| (i) | Explain why $f(x)$ is an increasing function for all x except one value.
Find this value and state what happens on the curve at this value. | 3 |
| (ii) | Hence show that $e^x \geq 1 + x^2$ for $x \geq 0$. | 2 |

Question 14 (15 marks) Start a NEW booklet.**Marks**

(a) (i) If ω is a non-real cube root of unity show that $1 + \omega + \omega^2 = 0$. 1

(ii) Hence evaluate $(1 - 3\omega + \omega^2)(1 + \omega - 8\omega^2)$. 2

(b) It is given that a , b and c are any real numbers.

(i) Prove that $a^2 + b^2 + c^2 \geq ab + bc + ca$. 2

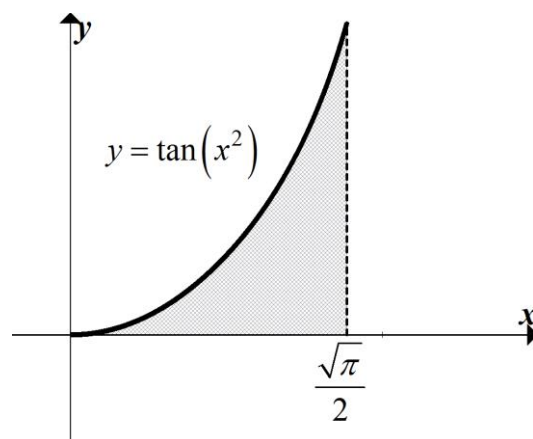
(ii) Hence or otherwise, prove that $3(a^2 + b^2 + c^2) \geq (a + b + c)^2$. 2

(c) (i) Find $\frac{d}{dx} [\ln(\cos x^2)]$. 1

(ii) The region in the plane which is bounded by the curve $y = \tan(x^2)$,

the x -axis and the line $x = \frac{\sqrt{\pi}}{2}$, shown in the diagram below,

is rotated about the y -axis to produce a solid S .



By using the method of cylindrical shells, find the volume of S in exact form. 3

Question 14 continues on page 11

Question 14 (continued)

Marks

(d) Let $I_n = \int_0^1 (1 - x^2)^n dx$ for $n \geq 1$.

(i) Show that $I_n = \frac{2n}{2n+1} I_{n-1}$

3

(ii) Hence evaluate I_2 .

1**End of Question 14**

Question 15 (15 marks) Start a NEW booklet.**Marks**

(a) (i) Show that $\int_0^a f(x) dx = \int_0^a f(a-x) dx$. 2

(ii) Hence deduce that $\int_0^{\frac{\pi}{4}} \frac{1 - \sin 2x}{1 + \sin 2x} dx = \int_0^{\frac{\pi}{4}} \tan^2 x dx$ 2

(iii) Hence evaluate $\int_0^{\frac{\pi}{4}} \frac{1 - \sin 2x}{1 + \sin 2x} dx$. 2

(b) $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$ where $p, q > 0$ are two distinct points

on the Hyperbola, H , $xy = c^2$.

(i) Show that the equation of the tangent to H at P is given by 2
 $x + p^2y = 2cp$.

(ii) The tangents to H at P and Q meet at T . 2
 Find the coordinates of T .

(iii) Show that the equation of the chord PQ is given by 2
 $x + pqy = c(p + q)$.

(iv) The chord PQ passes through the point $(2c, 0)$. 1
 Find the relationship between p and q .

(v) Hence, find the equation of the locus of T . 2
 For full marks a complete description must be given.

Question 16 (15 marks) Start a NEW booklet.**Marks**

a) Let $P(z) = 2z^3 - 5z^2 + qz - 5$, where q is a real number.

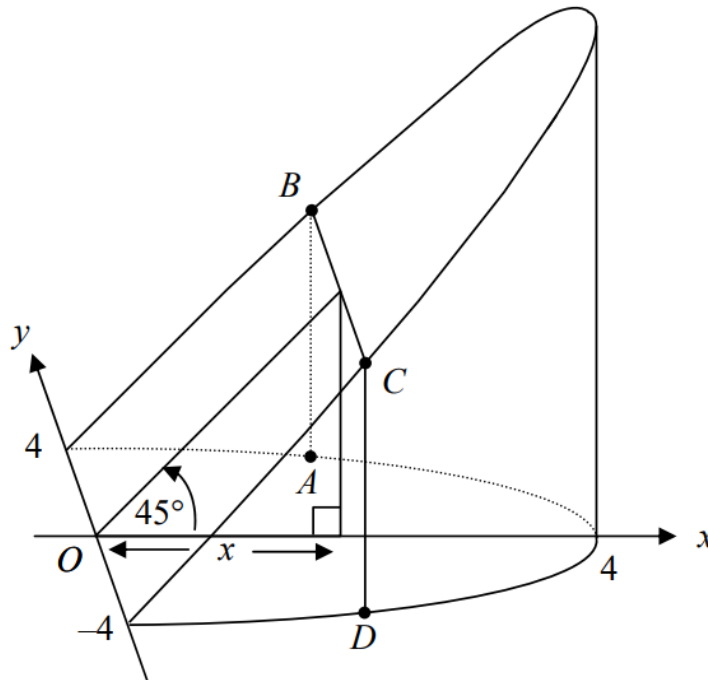
(i) If $P(1 - 2i) = 0$, solve $P(z) = 0$.

2

(ii) Hence determine the value of q if $P(1 - 2i) = 0$.

1

b) Lachlan's sculpture, "The Wedge", was obtained by cutting a right cylinder of radius 4 units at 45° through a diameter of its base as shown below.



A rectangular slice $ABCD$, of thickness δx , is taken perpendicular to the base of the wedge at a distance x from the y -axis.

(i) Show that the area of $ABCD$ is given by $2x\sqrt{16 - x^2}$.

2

(ii) Find the exact volume of the wedge.

3**Question 16 continues on page 14**

Question 16 (continued)

Marks

- (c) The equation $x^2 - x + 1 = 0$ has roots α and β , and $A_n = \alpha^n + \beta^n$ for $n \geq 1$.
- (i) Without solving the equation, show that $A_1 = 1$ and $A_2 = -1$. 2
- (ii) Show that $A_n = A_{n-1} - A_{n-2}$ for $n \geq 3$. 2
- (iii) Use mathematical induction to prove that $A_n = 2 \cos \frac{n\pi}{3}$ for $n \geq 1$. 3

END OF EXAM

Candidate Name/Number: _____

Multiple choice answer page. Fill in either A, B, C or D for questions 1-10.

This page must be handed in with your answer booklets

1.	
2.	
3.	
4.	
5.	

6.	
7.	
8.	
9.	
10.	

CHS YEAR 12 EXTENSION 2 2019	TRIAL HSC SUGGESTED SOLUTIONS
Multiple Choice Section: 1.C 2.A 3.B 4.C 5.B 6.D 7.D 8.A 9.B 10.A Question 1. $(1+2i)(-4-3i) = 2-11i$ ----- [C] Question 2. $B - A = C - B \rightarrow 2B = A + C$ $\frac{\sin(A+C)}{\sin B} = \frac{\sin 2B}{\sin B}$ $= \frac{2\sin B \cos B}{\sin B} = 2\cos B$ ----- [A] Question 3. $e = \frac{4}{5}, a = 10 \rightarrow ae = 8$ $\therefore -ae$ to $ae = 16$ ----- [B] Question 4. Since α, β, γ satisfy $x^3 - 3x^2 + 2 = 0$ Then $\alpha^3 = 3\alpha^2 - 2$ $\beta^3 = 3\beta^2 - 2$ $\gamma^3 = 3\gamma^2 - 2$ Hence $\alpha^3 + \beta^3 + \gamma^3 = 3(\alpha^2 + \beta^2 + \gamma^2) - 6$ $= 3(\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha) - 6$ $= 3(3)^2 - 2(0) - 6$ $= 21$ ----- [C] Question 5. $\frac{x}{y} + \frac{y}{x} = 2 \rightarrow (x-y)^2 = 0 \ [x \neq 0, y \neq 0]$ $\therefore y = x$ ----- [B]	Question 6. $x = 1 - \cos \theta, \ y = \theta - \sin \theta$ $\frac{dx}{d\theta} = \sin \theta, \ \frac{dy}{d\theta} = 1 - \cos \theta$ $\therefore \frac{dy}{dx} = \frac{1 - \cos \theta}{\sin \theta}$ $= \frac{2\sin^2 \frac{\theta}{2}}{2\sin \frac{\theta}{2} \cos \frac{\theta}{2}} = \tan \frac{\theta}{2}$ ----- [D] Question 7. $\arg(z - \omega)$ is positive (scale drawing) ----- [D] Question 8. $-1 \leq \frac{x-a}{b} \leq 1$ $\therefore -b \leq x-a \leq b$ $\therefore a-b \leq x \leq a+b$ ----- [A] Question 9. $\angle BAO + \angle AOC = \angle ABC + \angle OCB$ (exterior angle) $\angle AOC = 2\angle ABC$ ($\angle$ at centre = twice $\angle$ at circum) $\therefore 2x + 20 = x + 52 \rightarrow x = 32^\circ$ ----- [B] Question 10. $a = \frac{d}{dx} \left(\frac{1}{2} v^2 \right) = \frac{d}{dx} \left(\frac{1}{2} (\sin^{-1} x)^2 \right)$ $= \frac{1}{2} \times 2\sin^{-1} x \times \frac{1}{\sqrt{1-x^2}}$ ----- [A]

Question 11

a) i) $2\text{cis}\left(\frac{\pi}{6}\right)$

ii) $\left[2\text{cis}\left(\frac{\pi}{6}\right)\right]^4 = 16\text{cis}\left(\frac{2\pi}{3}\right) [DMT]$

$$= 16\left(-\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) = -8 + i8\sqrt{3}$$

b) Let $z = x + iy$.

$$\therefore |(x+2) + i(y-3)| = |(x-1) + iy|$$

$$\therefore x^2 - 2x + 1 + y^2 = x^2 + 4x + 4 + y^2 - 6y + 9$$

$$\therefore x - y + 2 = 0$$

c) $\int \sin \theta \cos^5 \theta d\theta$

$$\text{Let } u = \cos \theta \rightarrow du = -\sin \theta d\theta$$

$$\therefore I = -\int u^5 du = -\frac{u^6}{6}$$

$$\therefore I = -\frac{\cos^6 \theta}{6} + c$$

d) i)

$$\text{Let } 5 = A(x^2 + 4) + (Bx + C)(x+1) \text{ -----} \boxed{*}$$

$$\text{Let } x = -1 \text{ in } *: 5 = 5A \rightarrow A = 1$$

$$\text{Let } x = 0 \text{ in } *: 5 = 4A + C \rightarrow C = 1$$

$$\text{Let } x = 1 \text{ in } *: 5 = 5A + 2B + 2C \rightarrow B = -1$$

ii) $\int \frac{5}{(x+1)(x^2+4)} dx = \int \frac{1}{x+1} + \frac{1-x}{x^2+4} dx$

$$= \ln(x+1) - \frac{1}{2} \ln(x^2+4) + \frac{1}{2} \tan^{-1}\left(\frac{x}{2}\right) + c$$

$$= \ln\left(\frac{x+1}{\sqrt{x^2+4}}\right) + \frac{1}{2} \tan^{-1}\left(\frac{x}{2}\right) + c$$

e) $P(x) = x^3 + 3x^2 - 24x + k$

$$P'(x) = 3x^2 + 6x - 24$$

$$= 3(x^2 + 2x - 8)$$

$$= 3(x+4)(x-2)$$

$$\therefore P'(x) = 0 \rightarrow x = -4 \text{ or } 2$$

$$P(-4) = 80 + k = 0 \rightarrow k = -80$$

$$P(2) = k - 28 = 0 \rightarrow k = 28$$

Question 12.

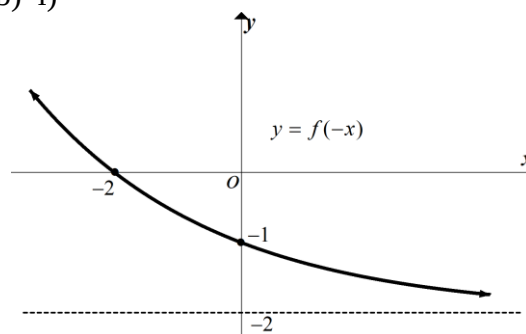
a) $\int \frac{x^2}{x+1} dx = \int \frac{x^2 - 1 + 1}{x+1} dx$

$$= \int \frac{(x-1)(x+1) + 1}{x+1} dx$$

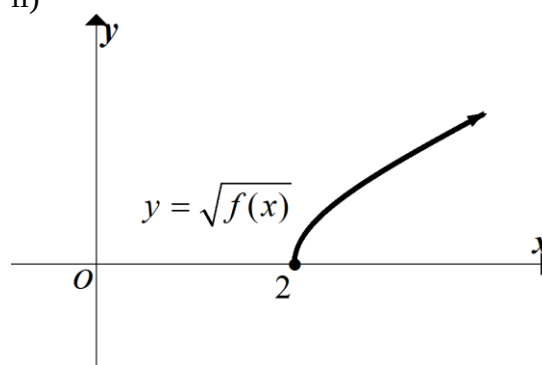
$$= \int x - 1 + \frac{1}{x+1} dx$$

$$= \frac{x^2}{2} - x + \ln(x+1) + c$$

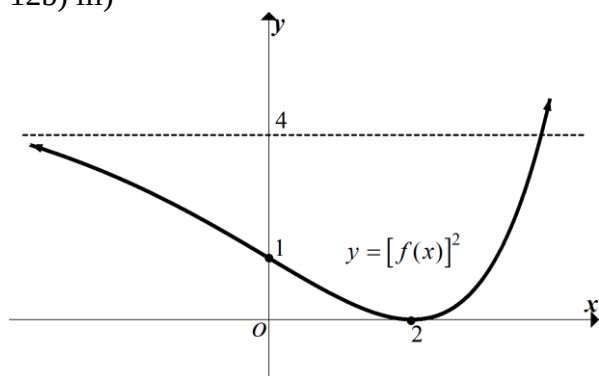
b) i)



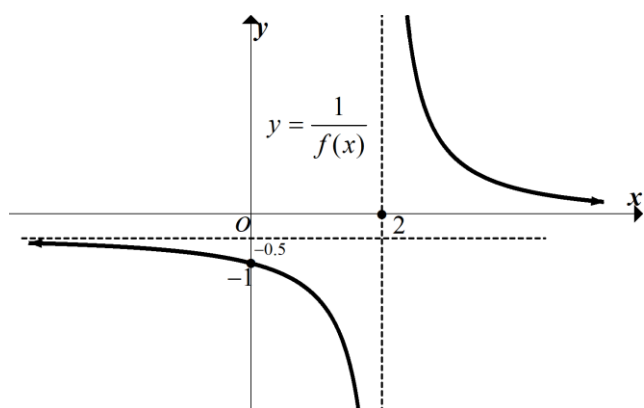
ii)



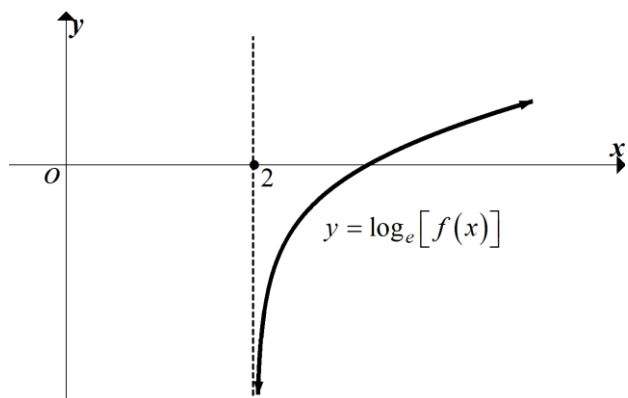
12b) iii)



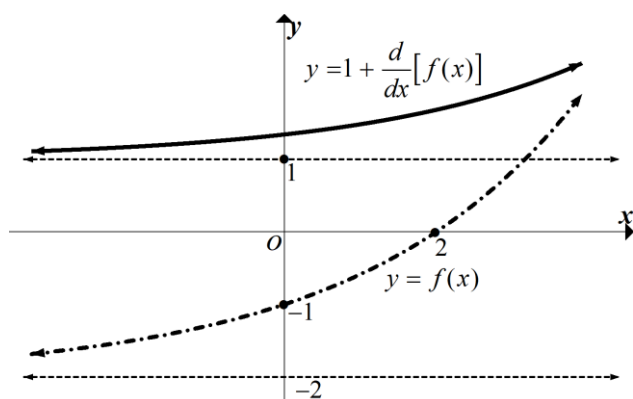
iv)



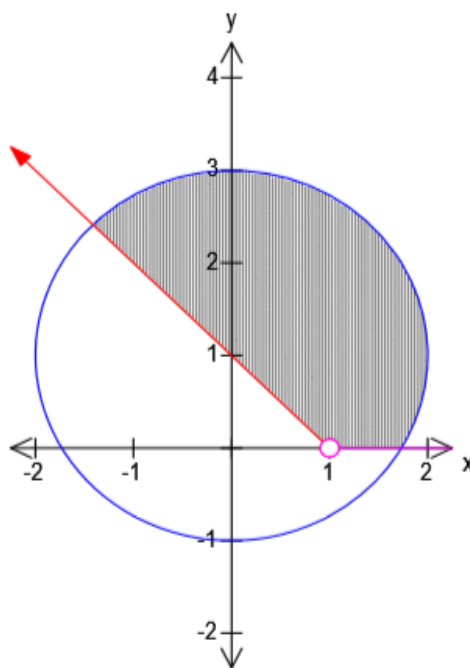
v)



vi)



c)



Question 13.

a) $x^2 - 9y^2 = 1 \rightarrow x^2 - 9y^2 = 1$

i) $b^2 = a^2(e^2 - 1) \rightarrow \frac{1}{9} = e^2 - 1$

$\therefore e = \frac{\sqrt{10}}{3}$

ii) Foci $\left(\pm \frac{\sqrt{10}}{3}, 0\right)$

iii) directrices $\therefore x = \pm \frac{3}{\sqrt{10}}$

b i) $\frac{x^2}{9} + \frac{y^2}{4} = 1 \rightarrow \frac{2x}{9} + \frac{2yy'}{4} = 0$

$\therefore y' = -\frac{4x}{9y}$ [gradient of tangent]

hence gradient of the normal is $\frac{9y}{4x}$

$\therefore$ at P: $m_N = \frac{9 \times 2 \sin \alpha}{4 \times 3 \cos \alpha} = \frac{3 \sin \alpha}{2 \cos \alpha}$

ii) Similarly the gradient of the normal at Q is

$m_N = \frac{3 \sin \beta}{2 \cos \beta}$

and as the normals are at right angles then:

$$\frac{3\sin \alpha}{2\cos \alpha} \times \frac{3\sin \beta}{2\cos \beta} = -1$$

$$\therefore \frac{3}{2} \tan \alpha \times \frac{3}{2} \tan \beta = -1$$

$$\therefore \cot \alpha \cot \beta = -\frac{9}{4}$$

c) The required roots are $\alpha\beta$, $\beta\gamma$ and $\alpha\gamma$
which is the same as $\frac{\alpha\beta\gamma}{\gamma}$, $\frac{\alpha\beta\gamma}{\alpha}$ and $\frac{\alpha\beta\gamma}{\beta}$.

Using sum of the roots $\alpha\beta\gamma = -1$, hence the required

$$\text{roots are } \frac{-1}{\alpha}, \frac{-1}{\beta} \text{ and } \frac{-1}{\gamma}.$$

$$\text{Let } x = \frac{-1}{\alpha} \rightarrow \alpha = \frac{-1}{x}$$

$\therefore$ the required equation is given by

$$\therefore \left(\frac{-1}{x}\right)^3 + 6\left(\frac{-1}{x}\right) + 1 = 0$$

$$\therefore \frac{-1}{x^3} - \frac{6}{x} + 1 = 0$$

$$\therefore x^3 - 6x^2 - 1 = 0$$

$$\text{c) } f(x) = x - \ln(1 + x^2)$$

$$\text{i) } f'(x) = 1 - \frac{2x}{1 + x^2}$$

$$f'(x) = \frac{1 + x^2 - 2x}{1 + x^2} = \frac{(x-1)^2}{1 + x^2}$$

$$\text{hence } f'(x) = \frac{(x-1)^2}{1 + x^2} \geq 0 \text{ thus is an increasing}$$

function for all x except $x=1$ where it is stationary.

$$\text{ii) Since } f(0) = 0 - \ln 1 = 0$$

$$\text{then } x - \ln(1 + x^2) \geq 0 \text{ for } x \geq 0.$$

$$\therefore x \geq \ln(1 + x^2)$$

$$\therefore e^x \geq e^{\ln(1+x^2)}$$

$$e^x \geq 1 + x^2$$

Question 14.

a) i) Since ω is a non-real cube root of unity then it satisfies $z^3 = 1$ but $\omega \neq 1$.

$$\therefore \omega^3 = 1 \rightarrow (\omega - 1)(\omega^2 + \omega + 1) = 0$$

$$\therefore \omega^2 + \omega + 1 = 0$$

ii)

$$(1 - 3\omega + \omega^2)(1 + \omega - 8\omega^2) = (-\omega - 3\omega)(-\omega^2 - 8\omega^2)$$

$$= (-4\omega)(-9\omega^2)$$

$$= 36\omega^3 = 36.$$

$$\text{b) i) } (a - b)^2 \geq 0$$

$$\therefore a^2 - 2ab + b^2 \geq 0$$

$$\therefore a^2 + b^2 \geq 2ab$$

similarly

$$\therefore b^2 + c^2 \geq 2bc \text{ and } \therefore c^2 + a^2 \geq 2ca$$

hence on adding

$$\therefore 2(a^2 + b^2 + c^2) \geq 2(ab + bc + ca)$$

$$\therefore a^2 + b^2 + c^2 \geq ab + bc + ca.$$

ii) Using (i)

$$2(a^2 + b^2 + c^2) \geq 2ab + 2bc + 2ca.$$

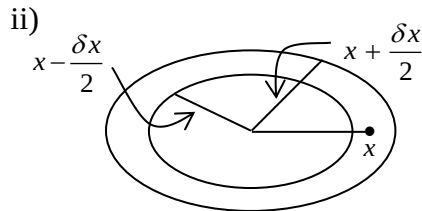
$$\therefore 3(a^2 + b^2 + c^2) \geq a^2 + b^2 + c^2 + 2ab + 2bc + 2ca.$$

and since

$$a^2 + b^2 + c^2 + 2ab + 2bc + 2ca = (a + b + c)^2$$

$$\text{then } 3(a^2 + b^2 + c^2) \geq (a + b + c)^2.$$

$$\begin{aligned} \text{c) i) } \frac{d}{dx} \left[\ln(\cos x^2) \right] &= \frac{-2x \sin(x^2)}{\cos(x^2)} \\ &= -2x \tan(x^2) \end{aligned}$$



Take a typical shell at a distance x units from O .

Thus the area of the annulus is given by

$$\begin{aligned} A &= \pi \left(x + \frac{\delta x}{2} + x - \frac{\delta x}{2} \right) \left(x + \frac{\delta x}{2} - \left(x - \frac{\delta x}{2} \right) \right) \\ &= \pi (2x)(\delta x) \\ &= 2\pi x \delta x \end{aligned}$$

$$\therefore \text{Volume of the shell} = 2\pi x \delta x y$$

$$\begin{aligned} \text{Total Volume} &= \lim_{\delta x \rightarrow 0} \sum_{x=0}^{\frac{\sqrt{\pi}}{2}} 2\pi x y \delta x \\ &= 2\pi \int_0^{\frac{\sqrt{\pi}}{2}} x \tan(x^2) dx \\ &= -\pi \left[\ln(\cos x^2) \right]_0^{\frac{\sqrt{\pi}}{2}} \\ &= -\pi \left[\ln\left(\cos \frac{\pi}{4}\right) - \ln(\cos 0) \right] \\ &= -\pi \left(\ln \frac{1}{\sqrt{2}} - \ln 1 \right) \\ &= \pi \ln(\sqrt{2}) = \frac{\pi}{2} \ln 2 \end{aligned}$$

$$\text{d) i) } I_n = \int_0^1 (1-x^2)^n dx \text{ for } n \geq 1.$$

$$\begin{aligned} \text{Using IBP: Let } u &= (1-x^2)^n, \quad v' = 1 \\ u' &= -2nx(1-x^2)^{n-1}, \quad v = x \end{aligned}$$

$$\begin{aligned} \therefore I_n &= x(1-x^2)^n \Big|_0^1 + \int_0^1 2nx^2(1-x^2)^{n-1} dx \\ &= 0 - 2n \int_0^1 ((1-x^2)-1)(1-x^2)^{n-1} dx \\ &= -2n \int_0^1 (1-x^2)^n - (1-x^2)^{n-1} dx \end{aligned}$$

$$\therefore I_n = -2n [I_n - I_{n-1}]$$

$$\therefore I_n + 2nI_n = 2nI_{n-1}$$

$$\therefore I_n(1+2n) = 2nI_{n-1}$$

$$I_n = \frac{2n}{2n+1} I_{n-1}$$

$$\text{ii) } I_2 = \frac{4}{5} I_1$$

$$\begin{aligned} I_1 &= \int_0^1 1-x^2 dx \\ &= x - \frac{x^3}{3} \Big|_0^1 = \frac{2}{3} \end{aligned}$$

$$\therefore I_2 = \frac{4}{5} \times \frac{2}{3} = \frac{8}{15}$$

Question 15.

$$\text{a) i) Let } I = \int_0^a f(x) dx \text{ and let } u = a - x$$

$$\therefore x = a - u \rightarrow dx = -du$$

$$\text{When } x = 0, u = a; \quad x = a, u = 0$$

$$\begin{aligned} \therefore I &= \int_a^0 f(a-u) \cdot (-du) \\ &= \int_0^a f(a-u) du = \int_0^a f(a-x) dx \end{aligned}$$

ii) Using (i)

$$\begin{aligned}\int_0^{\frac{\pi}{4}} \frac{1 - \sin 2x}{1 + \sin 2x} dx &= \int_0^{\frac{\pi}{4}} \frac{1 - \sin 2\left(\frac{\pi}{4} - x\right)}{1 + \sin 2\left(\frac{\pi}{4} - x\right)} dx \\&= \int_0^{\frac{\pi}{4}} \frac{1 - \sin\left(\frac{\pi}{2} - 2x\right)}{1 + \sin\left(\frac{\pi}{2} - 2x\right)} dx \\&= \int_0^{\frac{\pi}{4}} \frac{1 - \cos 2x}{1 + \cos 2x} dx \\&= \int_0^{\frac{\pi}{4}} \frac{1 - (1 - 2\sin^2 x)}{1 + (2\cos^2 x - 1)} dx \\&= \int_0^{\frac{\pi}{4}} \frac{2\sin^2 x}{2\cos^2 x} dx = \int_0^{\frac{\pi}{4}} \tan^2 x dx\end{aligned}$$

$$\begin{aligned}\text{iii) } \int_0^{\frac{\pi}{4}} \tan^2 x dx &= \int_0^{\frac{\pi}{4}} \sec^2 x - 1 dx \\&= [\tan x - x]_0^{\frac{\pi}{4}} = 1 - \frac{\pi}{4}\end{aligned}$$

$$\text{b) i) } y = \frac{c^2}{x} \rightarrow \frac{dy}{dx} = -\frac{c^2}{x^2}$$

$$\text{at } x = cp, \frac{dy}{dx} = m = -\frac{1}{p^2}$$

$$\therefore y - \frac{c}{p} = -\frac{1}{p^2}(x - cp)$$

$$\therefore p^2 y - cp = -x + cp$$

$$\therefore x + p^2 y = 2cp \text{ ----- [1]}$$

$$\text{ii) Similarly } x + q^2 y = 2cq \text{ ----- [2]}$$

$$\text{[2]} - \text{[1]}: (p^2 - q^2)y = 2c(p - q)$$

And since $p \neq q$

$$(p + q)y = 2c \rightarrow y = \frac{2c}{p + q}$$

$$\text{Subbing into [1] gives } x = \frac{2cpq}{p + q}$$

$$\therefore T\left(\frac{2cpq}{p + q}, \frac{2c}{p + q}\right)$$

$$\text{iii) } m_{PQ} = \frac{\frac{c}{q} - \frac{c}{p}}{cq - cp}$$

$$\begin{aligned}&\frac{c(p - q)}{c(q - p)} = -\frac{1}{pq}\end{aligned}$$

$$\therefore y - \frac{c}{p} = -\frac{1}{pq}(x - cp)$$

$$pqy - cq = -x + cp$$

$$\therefore x + pqy = c(p + q)$$

$$\text{iv) Using (iii) } 2c + 0 = c(p + q)$$

$$\therefore p + q = 2$$

$$\text{v) Using (iv) } T \text{ becomes } \left(\frac{2cpq}{2}, \frac{2c}{2}\right) \rightarrow T(cpq, c)$$

Hence the locus of T is $y = c$ for $0 < x < c$

Note: $p + q = 2$, both $p, q > 0$ and $p \neq q$

$$\therefore pq < 1$$

Also the tangents must meet outside of H in the first quadrant.

Also when $y = c$, $x = c$.

Question 16.

a) i) Since the coefficients are real, the roots occur in complex conjugate pairs.

Using sum of the roots:

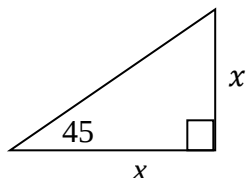
$$1 + 2i + 1 - 2i + \alpha = 0 \rightarrow \alpha = \frac{1}{2}$$

Hence the roots are $z = 1 + 2i$, $z = 1 - 2i$, $z = \frac{1}{2}$

ii) Since $z = \frac{1}{2}$ is a solution then $P\left(\frac{1}{2}\right) = 0$.

$$\therefore \frac{1}{4} - \frac{5}{4} + \frac{q}{2} - 5 = 0 \rightarrow q = 12$$

b) i)



Area of $ABCD = x \times 2y$

$$= x \times 2y$$

The base is half of the circle centre $(0,0)$, radius 4

Hence has equation

$$x^2 + y^2 = 4^2 \rightarrow y = \sqrt{16 - x^2}$$

$$\therefore \text{Area of } ABCD = x \times 2\sqrt{16 - x^2} \\ = 2x\sqrt{16 - x^2}$$

ii)

$$V_s = x \times 2\sqrt{16 - x^2} \delta x$$

$$\therefore V = \lim_{\delta x \rightarrow 0} \sum_{x=-4}^4 2x\sqrt{16 - x^2} \delta x$$

$$= 2 \int_{-4}^4 x\sqrt{16 - x^2} dx$$

$$\text{Let } u = 16 - x^2 \rightarrow du = -2x dx$$

$$\text{When } x = 0, u = 16; x = 4, u = 0$$

$$\therefore V = - \int_{16}^0 u^{\frac{1}{2}} du$$

$$= \int_0^{16} u^{\frac{1}{2}} du$$

$$= \left[\frac{2}{3} u^{\frac{3}{2}} \right]_0^{16}$$

$$= \frac{128}{3} u^{\frac{3}{2}}$$

c) i) $A_1 = \alpha^1 + \beta^1 = \text{sum of roots}$

$$= -\frac{b}{a} = -\left(\frac{-1}{1}\right) = 1$$

$$A_2 = \alpha^2 + \beta^2$$

$$= (\alpha + \beta)^2 - 2\alpha\beta$$

$$= 1^2 - 2 \times 1 = -1$$

ii) Since α, β satisfy $x^2 - x + 1 = 0$

$$\text{then } \alpha^2 - \alpha + 1 = 0 \text{ ---- [1]}$$

$$\beta^2 - \beta + 1 = 0 \text{ ---- [2]}$$

$$\text{[1]} \times \alpha^{n-2} \rightarrow \alpha^n - \alpha^{n-1} + \alpha^{n-2} = 0$$

$$\therefore \alpha^n = \alpha^{n-1} - \alpha^{n-2} \text{ ---- [3]}$$

$$\text{[2]} \times \beta^{n-2} \rightarrow \beta^n = \beta^{n-1} - \beta^{n-2} \text{ ---- [4]}$$

[3] + [4] gives:

$$\alpha^n + \beta^n = (\alpha^{n-1} + \beta^{n-1}) - (\alpha^{n-2} + \beta^{n-2})$$

$$\therefore A_n = A_{n-1} - A_{n-2}$$

$$\text{c) } A_1 = 2 \cos \frac{\pi}{3} = 2 \times \frac{1}{2} = 1$$

$$A_1 = 2 \cos \frac{2\pi}{3} = 2 \times -\frac{1}{2} = -1$$

$\therefore$ true for $n = 1$ and $n = 2$.

Assume true for $n = k$ i.e. $A_k = 2 \cos \frac{k\pi}{3}$

and $n = k - 1$ i.e. $A_{k-1} = 2 \cos(k-1) \frac{\pi}{3}$

$$\therefore A_{k+1} = A_k - A_{k-1} \quad \{ \text{using (b)} \}$$

$$= 2 \cos \frac{k\pi}{3} - 2 \cos(k-1) \frac{\pi}{3}$$

$$= 2 \cos \frac{k\pi}{3} - 2 \cos \frac{k\pi}{3} \cos \frac{\pi}{3} - 2 \sin \frac{k\pi}{3} \sin \frac{\pi}{3}$$

$$= \cos \frac{k\pi}{3} - 2 \sin \frac{k\pi}{3} \sin \frac{\pi}{3}$$

$$= 2 \cos \frac{k\pi}{3} \cos \frac{\pi}{3} - 2 \sin \frac{k\pi}{3} \sin \frac{\pi}{3}$$

$$= 2 \cos(k+1) \frac{\pi}{3}$$

Therefore if true when $n = k$, $k - 1$ then it is also true when $n = k + 1$.

Hence by induction is true for all $n \geq 1$.